

Math 252 - Calculus II - Final

December 6, 2023

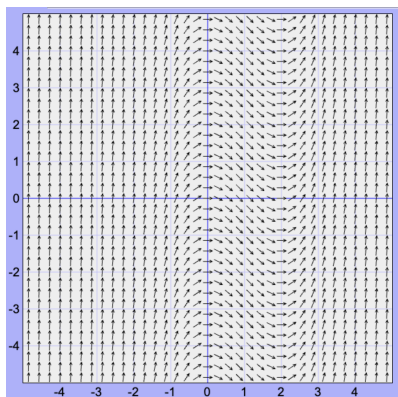
1. [3 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^2 - 2x + 4}{(x - 1)^2(x^2 + 4)^2} = \underline{\hspace{10cm}}$$

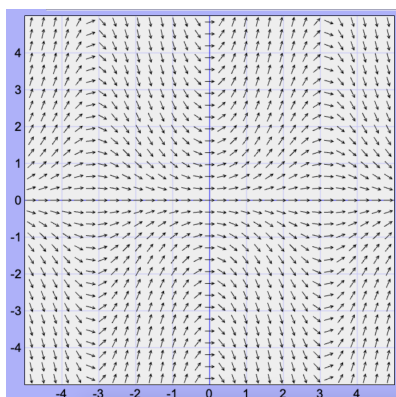
2. [5 pts] Match the differential equation with its direction field. Put A, B or C in the blanks.

- 1). Choose the direction field of $y' = y(x - 2)$. **Answer:** _____.
- 2). Choose the direction field of $y' = x^2 - 2x$. **Answer:** _____.
- 3). Choose the direction field of $y' = y \sin x$. **Answer:** _____.

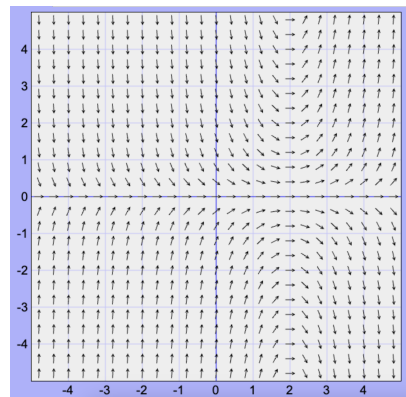
A



B



C



3. [3 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{x}{\sqrt[5]{x^8 - 0.2}} dx$ is **Convergent** or **Divergent** (circle the correct answer) by comparing it to the improper integral _____ (choose one of the following, and put A, B, C, or D in the blank).

A). $\int_1^{\infty} \frac{1}{x^5} dx$, B). $\int_1^{\infty} \frac{1}{x^8} dx$, C). $\int_1^{\infty} \frac{1}{x^{3/5}} dx$, D). $\int_1^{\infty} \frac{1}{x^{8/5}} dx$

4. [7 pts] Evaluate $\int \frac{x - 3x^4 e^{-x}}{x^4} dx$.

5. [7 pts] Evaluate $\int x \cos(2x) dx$.

6. [7 pts] Evaluate $\int \frac{2x-1}{x(x^2+1)} dx$.
7. [7 pts] Evaluate $\int \frac{x^2}{(1-x^2)^{3/2}} dx$.
8. [7 pts] Evaluate the improper integral $\int_2^\infty \frac{1}{x(\ln x)^2} dx$ or show that it is divergent.
9. [7 pts] Find the general solution to the differential equation

$$\frac{dy}{dx} = 4x\sqrt{y}.$$

Express your answer in the explicit function form of $y = f(x)$.

Answer: _____.

10. 1).[2 pts] Sketch the graph of the region bounded by $y = \frac{2}{x}$ and $y = 3 - x$. *Make sure to shade the region.*

2).[5 pts] Find the area of this region.

Answer: The area of the region is _____.

11. 1).[2 pts] Sketch the graph of the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{2}x$. *Make sure to shade the region.*

2).[5 pts] Find the volume of the solid obtained by rotating this region about x -axis.

Answer: The volume of the solid is _____.

12. 1).[2 pts] Sketch the graph of the region bounded by $y = x$, $y = \frac{1}{x}$ and $x = 3$ in the first quadrant. *Make sure to shade the region.*

2).[5 pts] Find the volume of the solid obtained by rotating this region about y -axis.

Answer: The volume of the solid is _____.

13. [7 pts] Find the average value of $f(x) = \sec^2 x \tan^3 x$ on the interval $[0, \frac{\pi}{3}]$.

Answer: The average value is _____.

14. [8 pts] A thin, two-meter bar, represented by the interval $0 \leq x \leq 2$, is made of an alloy whose density in units of kg/m is given by $\rho(x) = 2 + 3x^2$.

1). What is the mass of the bar?

2). What is the center of mass of the bar?

Answer: 1). The mass of the bar $m =$ _____ kg.

2). The center of mass of the bar $\bar{x} =$ _____ m.

15. [5 pts] A swimming pool has the shape of a rectangular box with a base that measures 20 meters by 30 meters and a uniform depth of 4 meters. Suppose the pool is filled with water to a depth of 3 meters. Set up a definite integral to find the work required to pump the water out over the top of the swimming pool.

The density of water is $\rho = 1000 \text{ kg/m}^3$, and the acceleration due to gravity is $g = 9.8 \text{ m/s}^2$.

Note: You do not need to simplify the integrand function or evaluate the integral.

Answer: Work = $\int_{\text{_____}}^{\text{_____}}$ _____ J.

16. [6 pts] The population of a rabbit community is governed by the following logistic equation with the initial population,

$$\frac{dP}{dt} = \frac{1}{5000}P(1000 - P), \quad P(0) = 250.$$

- 1). Find the population function $P(t)$, that is, the solution of the logistic equation.
2). Find $\lim_{t \rightarrow \infty} P(t)$.

Answer: 1). The population function $P(t) =$ _____.

- 2). $\lim_{t \rightarrow \infty} P(t) =$ _____.

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Last Name: _____,

First Name: _____

17. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - 2x^2 + x - 1}{x^2(x-1)(x^2+1)^2} = \underline{\hspace{10cm}}$$

18. [3 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{x^2}{\sqrt[3]{x^7 - 0.1}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x^2}{\sqrt[3]{x^7 - 0.1}}$ _____ (choose one between A and B , and then choose one among C, D, E , and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{7/3}}$, D). $\frac{1}{x^{1/3}}$, E). $\frac{1}{x^5}$, F). $\frac{1}{x^{3/7}}$

19. [8 pts] Evaluate $\int \tan^3 x \sec^4 x dx$.

20. [8 pts] Evaluate $\int x \sin(3x) dx$.

21. [8 pts] Evaluate $\int \frac{x+18}{x(x^2+9)} dx$.

22. [8 pts] Evaluate $\int \frac{1}{x^2\sqrt{x^2-4}} dx$.

23. [7 pts] Evaluate the improper integral $\int_1^{\infty} xe^{-x^2} dx$ or show that it is divergent.

(Note: Answers without appropriate limit notations will result in loss of points.)

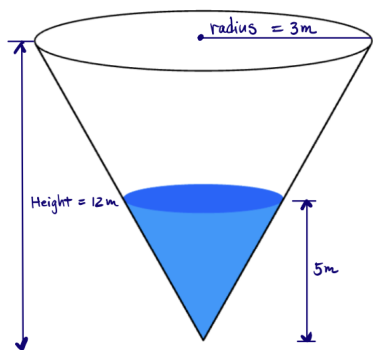
24. [7 pts] Solve the differential equation $\frac{dy}{dx} = y^2 \cos x$.

25. 1).[2 pts] Sketch the graph of the region bounded by $y = -x^2 + 5x$ and $y = x$. *Make sure to shade the region.*

2).[5 pts] Find the area of this region.

26. 1).[2 pts] Sketch the graph of the region bounded by $y = \sqrt{x}$ and $y = x$. *Make sure to shade the region.*

- 2).[5 pts] Find the volume of the solid obtained by rotating this region about x -axis.
27. 1).[2 pts] Sketch the graph of the region bounded by $y = x^2$, $y = \frac{1}{x}$ and $x = 2$. *Make sure to shade the region.*
- 2).[5 pts] Find the volume of the solid obtained by rotating this region about y -axis.
28. [7 pts] Find the average value of $f(x) = \frac{3x^3 - \sqrt{x}}{x}$ on the interval $[1, 9]$.
29. [8 pts] A thin, one-meter bar, represented by the interval $1 \leq x \leq 2$, is made of an alloy whose density in units of kg/m is given by $\rho(x) = 2 + \frac{6}{x^4}$.
- 1). What is the mass of the bar?
- 2). What is the center of mass of the bar?
- 2). The center of mass of the bar $\bar{x} = \underline{\hspace{2cm}}$ m.
30. [5 pts] An oil tank has a shape of inverted circular cone with height 12 meters and base radius 3 meters, as shown in the following figure. The tank is only filled with oil to a depth of 5 meters. Set up a definite integral to find the work required to pump the oil in the tank out over the top of the tank.
- The density of oil is $\rho = 800 \text{ kg/m}^3$, and the acceleration due to gravity is $g = 9.8 \text{ m/s}^2$.
Note: You do not need to simplify the integrand function or evaluate the integral.



Answer: Work = $\int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{4cm}} d\underline{\hspace{1cm}}$ J.

31. [5 pts] The fish population in a lake is modeled by the following logistic equation with initial condition,
- $$\frac{dP}{dt} = 0.04P \left(1 - \frac{P}{1000} \right), \quad P(0) = 200.$$
- 1). Find the population function $P(t)$, that is, the solution of the logistic equation.
- 2). Find $\lim_{t \rightarrow \infty} P(t)$.

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March 20, 2024 – Version 2

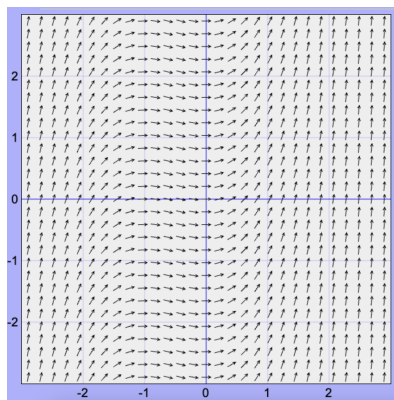
32. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - 2x^2 + x - 1}{x^2(x+1)(x^2+4)^2} = \underline{\hspace{10cm}}$$

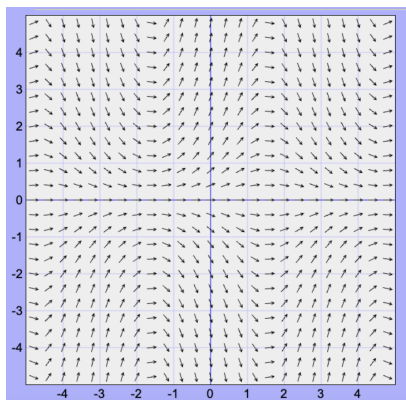
33. [3 pts] Match the differential equation with its direction field. Put A, B or C in the blanks.

- 1). Choose the direction field of $y' = x(y+2)$. **Answer:** _____.
- 2). Choose the direction field of $y' = x^2 + x$. **Answer:** _____.
- 3). Choose the direction field of $y' = y \cos x$. **Answer:** _____.

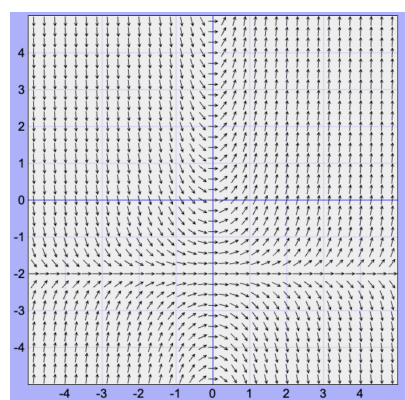
A



B



C



34. [3 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{x}{\sqrt[4]{x^9+1}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x}{\sqrt[4]{x^9+1}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

- A). $\geq$ B). $\leq$, C). $\frac{1}{x^{5/4}}$, D). $\frac{1}{x^{9/4}}$, E). $\frac{1}{x^{4/9}}$, F). $\frac{1}{x^{5/9}}$

35. [8 pts] Evaluate $\int \tan^3 x \sec^4 x \, dx$.
36. [8 pts] Evaluate $\int x \cos(3x) \, dx$.
37. [8 pts] Evaluate $\int \frac{x+8}{x(x^2+4)} \, dx$.
38. [8 pts] Evaluate $\int \frac{1}{x^2\sqrt{x^2-4}} \, dx$.
39. [7 pts] Evaluate the improper integral $\int_1^\infty xe^{-x^2} \, dx$ or show that it is divergent.

(Note: Answers without appropriate limit notations will result in loss of points.)

40. [7 pts] Solve the differential equation $\frac{dy}{dx} = y^2 \cos x$.

Express your answer in the explicit function form of $y = f(x)$.

Answer: _____.

41. 1).[2 pts] Sketch the graph of the region bounded by $y = -x^2 + 3x$ and $y = x$. *Make sure to shade the region.*

2).[5 pts] Find the area of this region.

Answer: The area of the region is _____.

42. 1).[2 pts] Sketch the graph of the region bounded by $y = \sqrt{x}$ and $y = x$. *Make sure to shade the region.*

2).[5 pts] Find the volume of the solid obtained by rotating this region about x -axis.

Answer: The volume of the solid is _____.

43. 1).[2 pts] Sketch the graph of the region bounded by $y = x^2$, $y = \frac{1}{x}$ and $x = 2$. *Make sure to shade the region.*

2).[5 pts] Find the volume of the solid obtained by rotating this region about y -axis.

Answer: The volume of the solid is _____.

44. [7 pts] Find the average value of $f(x) = \frac{4x^4 - \sqrt{x}}{x}$ on the interval $[1, 4]$.

Answer: The average value is _____.

45. [8 pts] A thin, one-meter bar, represented by the interval $1 \leq x \leq 2$, is made of an alloy whose density in units of kg/m is given by $\rho(x) = 2 + \frac{6}{x^4}$.

1). What is the mass of the bar?

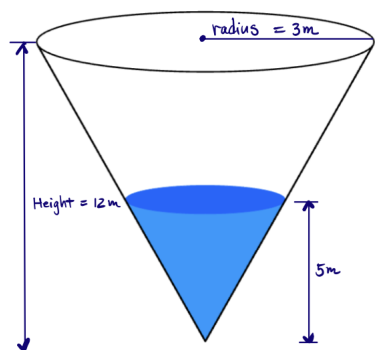
2). What is the center of mass of the bar?

Answer: 1). The mass of the bar $m =$ _____ kg.

2). The center of mass of the bar $\bar{x} =$ _____ m.

46. [5 pts] An oil tank has a shape of inverted circular cone with height 12 meters and base radius 3 meters, as shown in the following figure. The tank is only filled with oil to a depth of 5 meters. Set up a definite integral to find the work required to pump the oil in the tank out over the top of the tank.

The density of oil is $\rho = 800 \text{ kg/m}^3$, and the acceleration due to gravity is $g = 9.8 \text{ m/s}^2$.
Note: You do not need to simplify the integrand function or evaluate the integral.



Answer: Work = $\int_{\text{---}}^{\text{---}}$ _____ d _____ J.

47. [5 pts] The fish population in a lake is modeled by the following logistic equation with initial condition,

$$\frac{dP}{dt} = 0.06P \left(1 - \frac{P}{900} \right), \quad P(0) = 300.$$

- 1). Find the population function $P(t)$, that is, the solution of the logistic equation.
- 2). Find $\lim_{t \rightarrow \infty} P(t)$.

Answer: 1). The population function $P(t) =$ _____.

2). $\lim_{t \rightarrow \infty} P(t) =$ _____.

June 10, 2024

- $$\frac{3 + 2x - x^2 + 4x^3}{(x + 1)^3(x^2 + 4)^2} = \underline{\hspace{10cm}}$$

- The improper integral $\int_2^\infty \frac{x}{\sqrt[4]{x^7-1}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x}{\sqrt[4]{x^7-1}}$ _____ (choose one between A and B , and then choose one among C, D, E , and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{7/4}}$, D). $\frac{1}{x^6}$, E). $\frac{1}{x^{3/4}}$, F). $\frac{1}{x^3}$

50. [6 pts] Evaluate $\int \frac{2x - 4x(\sqrt[3]{x}) + 5x^2(\sqrt{x})}{x} dx$.

- 51.** [6 pts] Evaluate $\int \frac{(\arctan x)^2}{x^2 + 1} dx$.

- 52.** [6 pts] Evaluate $\int \frac{5x + 9}{x(2x + 3)} dx$.

- 53.** [6 pts] Evaluate $\int x e^{-2x} dx$.

54. [8 pts] Evaluate $\int \frac{1}{x^2 \sqrt{16 + x^2}} dx$.

55. [6 pts] Evaluate $\int \sec^3 \theta \tan^3 \theta d\theta$.

- 56.** [7 pts] Evaluate the improper integral $\int_2^{\infty} \frac{2}{x(\ln x)^3} dx$ or show that it is divergent.

57. [7 pts] Find the solution to the initial value problem

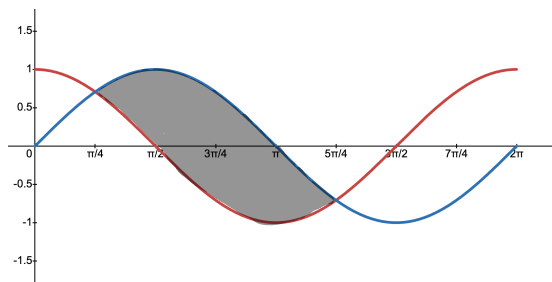
$$\frac{dy}{dx} = y^2 \sec x, \quad y(0) = -\frac{1}{3}.$$

Answer: _____.

58. [6 pts] Find the average value of $f(x) = \frac{x^2}{\sqrt{1+x^3}}$ on the interval $[0, 2]$.

Answer: The average value is _____.

59. [6 pts] The following shaded region is bounded by $y = \sin x$ and $y = \cos x$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$. Find the area of the shaded region.



Answer: The area of the shaded region is _____.

60. [6 pts] The force (in Newtons) on a particle is described by

$$F(x) = 3 \cos^2 x \sin x$$

at a point x (in meters) along the x -axis. Find the work done in moving the particle from $x = 0$ to $x = \pi$.

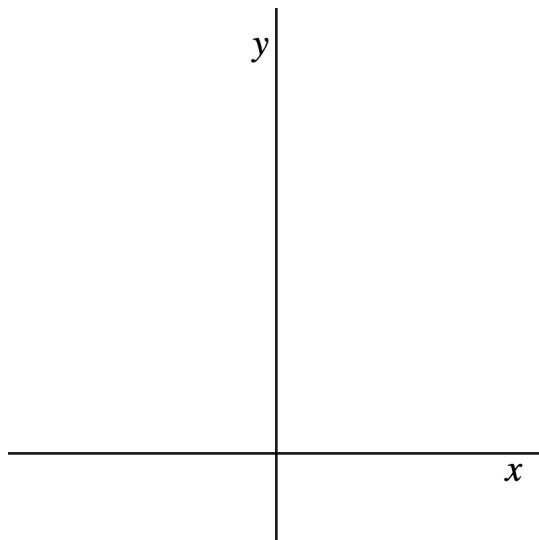
Answer: Work = _____ J.

61. 1).[2 pts] Sketch the graph of the region bounded by $y = \frac{8}{x}$, $y = \sqrt{x}$, and $x = 8$. *Make sure to shade the region.*
2).[5 pts] Find the volume of the solid obtained by rotating this region about x -axis.

Answer: The volume of the solid is _____.

62. Consider the region bounded by $y = 4 - x^2$ and $y = -2x + 4$. The volume of the solid obtained by rotating this region about **y -axis** can be computed by using both disk/washer and cylindrical shell methods.

- 1). [2 pts] Sketch the graph of the above region. *Make sure to shade the region.*



- 2). [6 pts] Set up, **but do not evaluate**, integrals to compute of the volume of the solid using both methods. Fill in the blanks in the answer lines to complete the setup of the integrals.

Note: You do not need to simplify the integrand function or evaluate the integral.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$$

ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$$

63. [6 pts] A thin metal bar is 2-meter long (starting at $x = 1$ meter), and has a density function of $\rho(x) = 2 + \frac{18}{x^3}$ kg/m.

1). Find the mass of the bar. 2). Find the center of mass of the bar.

Note: Simplify your answers whenever possible.

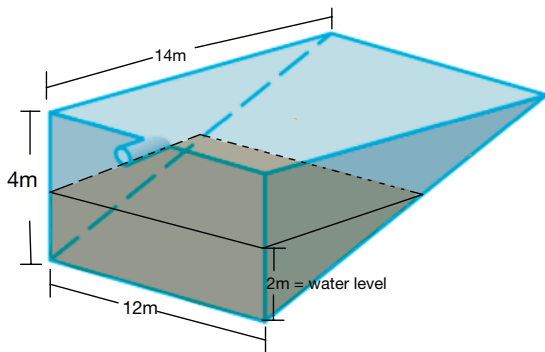
Answer: 1). The mass of the bar $m = \underline{\hspace{2cm}}$ kg.

2). The center of mass of the bar $\bar{x} = \underline{\hspace{2cm}}$ m.

64. [4 pts] A water tank has the following shape. The current water level is only 2 meters deep as shown in the graph. Set up a definite integral to find the work required to pump the water out of the spout, which is at the same level as the top of the tank.

The density of water is $\rho = 1000$ kg/m³, and the constant $g = 9.8$ m/s².

Note: You do not need to simplify the integrand function or evaluate the integral.



Answer: Work = $\int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$ J.

Math 252 - Calculus II - Final

December 9, 2024

65. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 + 2x + 1}{x(x-2)^3(x^2+1)^2} = \underline{\hspace{10cm}}$$

66. [2 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{x}{\sqrt[3]{x^8+3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x}{\sqrt[3]{x^8+3}} \underline{\hspace{1cm}} \underline{\hspace{1cm}}$ (choose one between A and B , and then choose one among C , D , E , and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{8/3}}$, D). $\frac{1}{x^{3/8}}$, E). $\frac{1}{x^{5/3}}$, F). $\frac{1}{x}$

67. [6 pts] Evaluate $\int \frac{x^4 - x^2\sqrt{x} + 2}{x^3} dx$.

68. [6 pts] Evaluate $\int_0^1 x^2 e^{-x^3} dx$.

69. [6 pts] Evaluate $\int \frac{\ln x}{x^2} dx$.

70. [8 pts] Use the method of partial fractions to evaluate $\int \frac{x+8}{x(x^2+4)} dx$.

Note: No credit will be given if you do not use partial fraction method.

71. [8 pts] Evaluate $\int \frac{x^2}{\sqrt{4-x^2}} dx$.

Note: Simplify your answer so that it does not contain any trigonometric composed with inverse trigonometric terms.

72. [8 pts] Find the average value of $f(x) = \tan^3 x \sec^4 x$ on the interval $[0, \frac{\pi}{4}]$.

73. [8 pts] Evaluate the improper integral $\int_{\sqrt{3}}^{\infty} \frac{\arctan x}{1+x^2} dx$ or show that it is divergent.

Note: Use the limit definition of improper integrals to compute the above integral. Do not use Comparison Test.

74. [8 pts] Find the solution to the initial value problem

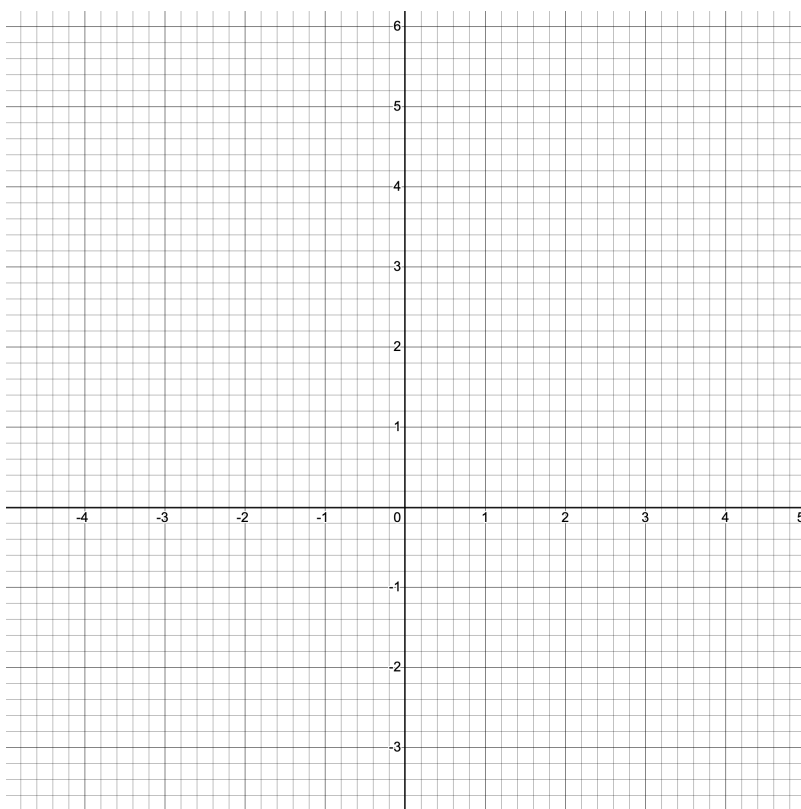
$$\frac{dy}{dx} = \frac{4x}{e^{2y}}, \quad y(1) = 0.$$

Express your answer in the explicit function form of $y = f(x)$.

75. [8 pts] A thin metal bar with a density function of $\rho(x) = 2 + \cos x$ (in kg/m) lies on the x -axis between $x = 0$ and $x = \frac{\pi}{2}$ (in meters).

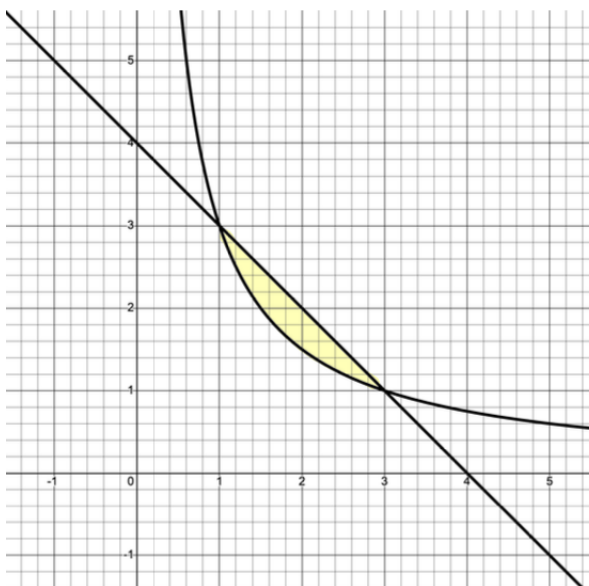
1). Find the mass of the bar. 2). Find the center of mass of the bar.

76. [2 pts] 1). Sketch the graph of the region bounded by $y = 4 - x^2$ and $y = -x + 2$, and find the coordinates of the intersection points of these two curves. *Make sure to shade the region.*



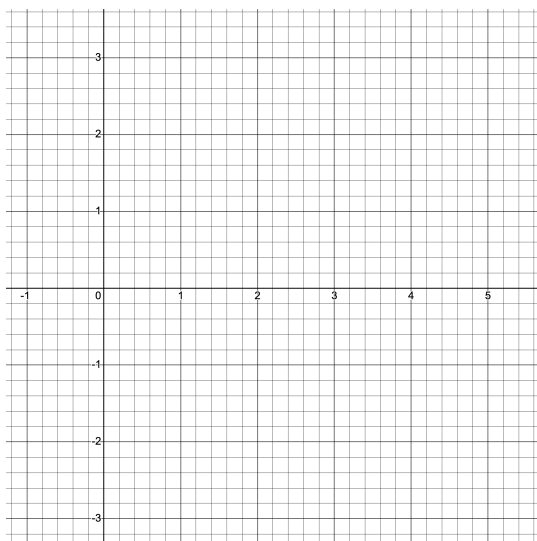
2). [5 pts] Find the **area** of the region in Part 1).

77. [7 pts] The following shaded region is bounded by $y = 4 - x$ and $y = \frac{3}{x}$. Find the **volume** of the solid obtained by rotating this region about y -axis.



78. Consider the region bounded by $y = \frac{x}{2}$ and $y = \sqrt{x}$. The volume of the solid obtained by rotating this region about the horizontal line $y = -1$ can be computed by using both disk/washer and cylindrical shell methods.

- 1). [2 pts] Sketch the graph of the region bounded by $y = \frac{x}{2}$ and $y = \sqrt{x}$, and find the coordinates of the intersection points of these two curves. *Make sure to shade the region.*



- 2). [6 pts] Set up, **but do not evaluate**, integrals to compute of the volume of the solid obtained by rotating the region in Part 1) about the horizontal line $y = -1$ using both methods.

Note: You do not need to simplify the integrand function or evaluate the integral.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{10em}} d\underline{\hspace{1em}}$$

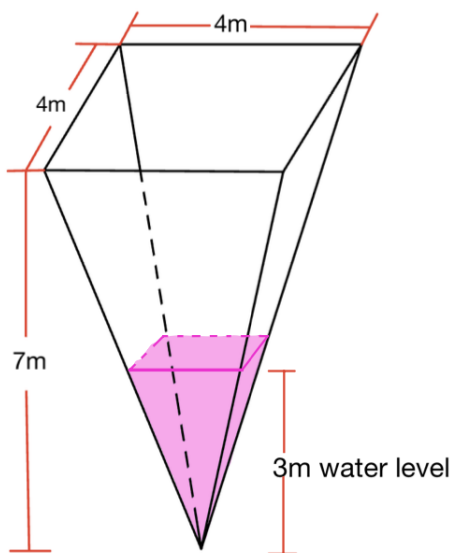
ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{10em}} d\underline{\hspace{1em}}$$

79. [4 pts] A water tank has the following inverted pyramid shape with a square top. The tank is 7 meter high and the square top has sides of 4 meters, although the current water level is only at 3 meter deep. Set up a definite integral to find the work required to pump the water out of the top of the tank.

The density of water is $\rho = 1000 \text{ kg/m}^3$, and the constant acceleration due to gravity is $g = 9.8 \text{ m/s}^2$.

Note: You do not need to simplify the integrand function or evaluate the integral.



Answer: Work = $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{10em}} d\underline{\hspace{1em}}$ Joules.

80. [4 pts] The fish population (in hundreds) in a lake is modeled by the following logistic equation with the initial population,

$$\frac{dP}{dt} = \frac{1}{30}P(6 - P), \quad P(0) = 4.$$

- 1). Find the population function $P(t)$, that is, the solution of the logistic equation.
- 2). Find $\lim_{t \rightarrow \infty} P(t)$.

Math 252 - Calculus II - Final Exam

CRN: 23526

March 18, 2025

81. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - x^2 + 2x + 1}{x(x+2)^3(x^2+4)^2} = \underline{\hspace{10cm}}$$

82. [2 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_2^{\infty} \frac{x^3}{\sqrt{x^7-1}} dx$ is _____ (choose “convergent” or “divergent” to fill in the blank) because $\frac{x^3}{\sqrt{x^7-1}}$ _____ (choose one between A and B , and then choose one among C, D, E , and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{7/2}}$, D). $\frac{1}{x^4}$, E). $\frac{1}{x^{1/2}}$, F). $\frac{1}{x^{3/7}}$

83. [5 pts] Evaluate $\int \frac{x^6 - x^3 + 3}{x^4} dx$.

84. [5 pts] Evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$.

85. [6 pts] Evaluate $\int x \sin(5x) dx$.

86. [6 pts] Evaluate $\int e^x \sin x dx$.

Hint: Use integration by parts twice.

87. [8 pts] Use the method of partial fractions to evaluate $\int \frac{x+1}{x(x^2+1)} dx$.

Note: No credit will be given if you do not use partial fraction method.

88. [8 pts] Use trigonometric substitution to evaluate $\int \frac{1}{x^2\sqrt{x^2-4}} dx$.

Note: Simplify your answer so that it does not contain any trigonometric composed with inverse trigonometric terms.

89. [8 pts] Find the average value of $f(x) = \sin^2 x \cos^3 x$ on the interval $[0, \frac{\pi}{6}]$.

Note: Simplify your trigonometric values, but you do not need to simplify fractions with distinct denominators if there is any.

90. [8 pts] Evaluate the improper integral $\int_2^{\infty} \frac{1}{x(\ln x)^2} dx$ or show that it is divergent.

Note: Use the limit definition of improper integrals to compute the above integral. Do not use Comparison Test.

91. [8 pts] Find the solution to the initial value problem

$$\frac{dy}{dx} = \frac{4x^3}{y^4}, \quad y(0) = 2.$$

Express your answer in the explicit function form of $y = f(x)$.

92. [7 pts] A thin metal bar with a density function of $\rho(x) = 6 + \frac{8}{x^3}$ (in kg/m) lies on the x -axis between $x = 1$ and $x = 2$ (in meters).

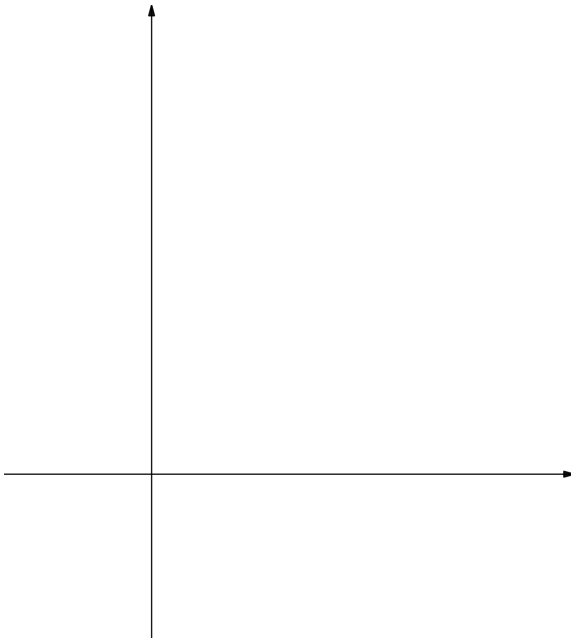
- 1). Find the mass of the bar.
- 2). Find the center of mass of the bar.

Note: Simplify your answers whenever possible.

Answer: 1). The mass of the bar $m = \underline{\hspace{2cm}}$ kg.

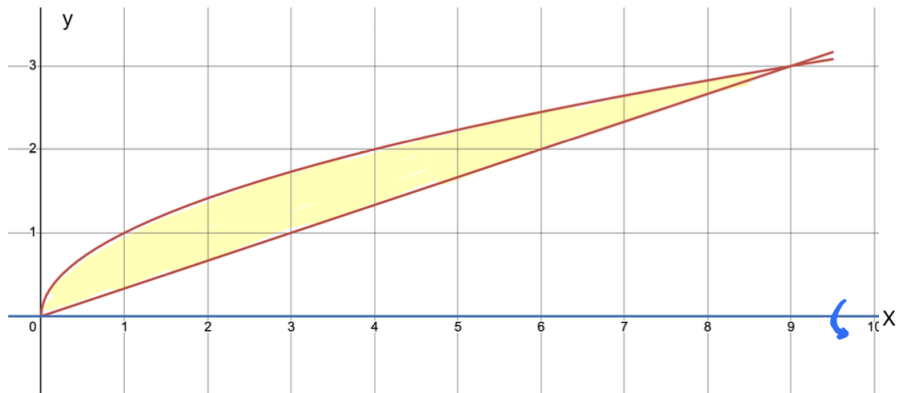
2). The center of mass of the bar $\bar{x} = \underline{\hspace{2cm}}$ m.

93. [2 pts] 1). Sketch the graph of the region bounded by $y = \frac{1}{x}$, $y = 4x$ and $x = 1$ in the first quadrant. Then find the coordinates of the intersection point of $y = \frac{1}{x}$ and $y = 4x$ in the first quadrant.

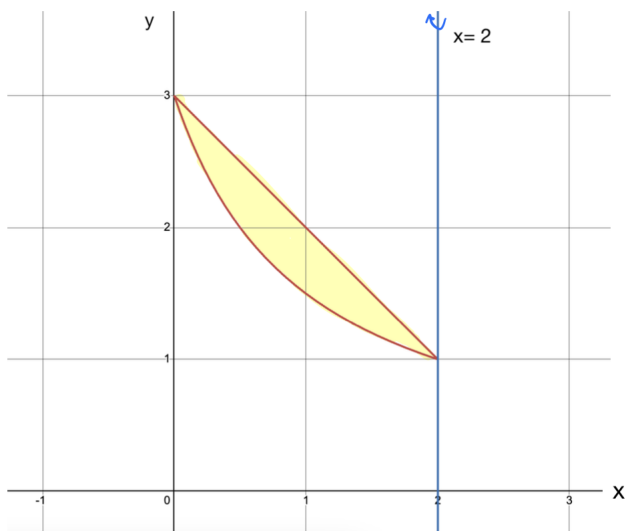


- 2). [5 pts] Find the **area** of the region in Part 1).

94. [6 pts] The following shaded region is bounded by $y = \frac{x}{3}$ and $y = \sqrt{x}$. Find the **volume** of the solid of revolution obtained by rotating this region about x -axis.



95. [6 pts] The following shaded region is bounded by $y = 3 - x$ and $y = \frac{3}{x+1}$.



Set up, **but do not evaluate**, definite integrals to compute of the volume of the solid obtained by rotating the above shaded region about **the vertical line** $x = 2$ using both disk/washer and cylindrical shell methods.

Fill in the blanks in the following answer line. You do not need to simplify the integrand functions or evaluate the integrals.

Hint: The curve $y = \frac{3}{x+1}$ is equivalent to $x = \frac{3}{y} - 1$. The line $y = 3 - x$ is the same as $x = 3 - y$.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$$

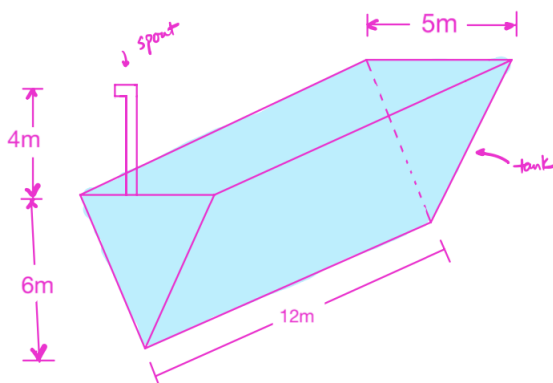
ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{10em}} d\underline{\hspace{1em}}$$

96. [4 pts] A tank in the following inverted triangular prism shape is *full of water*. Set up a definite integral to compute the work required to pump all the water out of the spout, which is 4m above the top of the tank as shown in the picture.

The density of the water is 1000 kg/m^3 , and the gravitational acceleration is 9.8 m/s^2 .

Fill in the blanks in the following answer line. **Do not evaluate the definite integral.**



Answer: Work = $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{10em}} d\underline{\hspace{1em}}$ J.

97. [4 pts] A population of squirrels (in hundreds) that lives in a forest is modeled by the following logistic equation with the initial population,

$$\frac{dP}{dt} = \frac{1}{150}P(30 - P), \quad P(0) = 6.$$

- 1). For what values of P is the population increasing? **Answer:** _____
- 2). For what values of P is the population decreasing? **Answer:** _____
- 3). Find the population function $P(t)$, that is, the solution to the logistic equation.

Answer: $P(t) =$ _____

- 4). Find $\lim_{t \rightarrow \infty} P(t)$. **Answer:** _____